

Basic Principles of Medicine 1
Module: Foundations to Medicine
Lecture 5

Resting Membrane Potentials

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Learning Objectives

- SOM.MK.II.BPM1.1.FTM.3.PHYS.0025 Predict the difference in free energy of a solute and solvent between two compartments, including chemical and electrical forces. Evaluate if the compartments are in a steady state and/or equilibrium.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0026 Based on the principle of ionic attraction, explain how a potential difference across a membrane will influence the distribution of a cation and an anion.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0027 Write the Nernst equation and indicate how this equation accounts for both the chemical and electrical driving forces that act on an ion. Explain the effects of altering the intracellular or extracellular Na^+ , K^+ , Cl^- , or Ca^{2+} concentration on the equilibrium potential for that ion.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0028 List the concentrations and equilibrium potentials of Na, K, Cl, Ca in a typical non-excitabile cell. Based on the Nernst equilibrium potential, predict the direction of transmembrane flux for an ion when the membrane potential a) is at its equilibrium potential, b) is higher than the equilibrium potential, or c) is less than the equilibrium potential.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0029 Be able to calculate the equilibrium potential for an ion using the Nernst equation. Contrast the difference in E_K (the Nernst potential for K^+) caused by a 5 mEq/l increase in extracellular K^+ with the change in E_{Na} (the Nernst potential for Na^+) caused by a 5 mEq/l increase in extracellular Na^+ .
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0030 Using the Goldman-Hodgkin-Katz equation, explain how the relative permeabilities of the 4 key ion species (Na^+ , K^+ , Cl^- , Ca^{2+}) create the resting membrane potential. Given an increase or decrease in the permeability for each of the ions, predict how the membrane potential would change.

Recommended Reading

Rhoades and Bell, Medical Physiology

- Chapter 3: Section 'Resting Membrane Potentials'

Abbreviations

Na-K ATPase – sodium-potassium ATPase (active pump)

mM – millimol (1/1000 mol)

mV – millivolt (1/1000 Volt)

E_{ion} – Equilibrium potential / Nernst potential / Diffusion potential for the specific ion

V_m – membrane potential (measured)

RMP – resting membrane potential

log – base 10 logarithm, i.e. $\log_{10}$

Ψ_{ion} – driving force for an ion = $V_m - E_{ion}$

Ions:

H^+ – hydrogen

K^+ – potassium

Na^+ – sodium

Cl^- – chloride

Outline

- Important principles
- Ohm's law, Resistance, Conductance and flux
- Ion-specific equilibrium potentials
- The Nernst equation
- Fractional conductance
- Resting potential and the Goldman-Hodgkin-Katz equation
- Effects of changes in ion concentrations

Ohm's Law and conductance

- Ohm's law:

$$I = V/R$$

- Current (I, in Ampere) = Driving force (Volt) / Resistance (Ohm)
- g (Conductance) = ease of flow
- Conductance and resistance are reciprocal: $1/R = g$
- Hence the inverse of conductance can be substituted for resistance:

$$I = V \times g$$

Electric current simplified

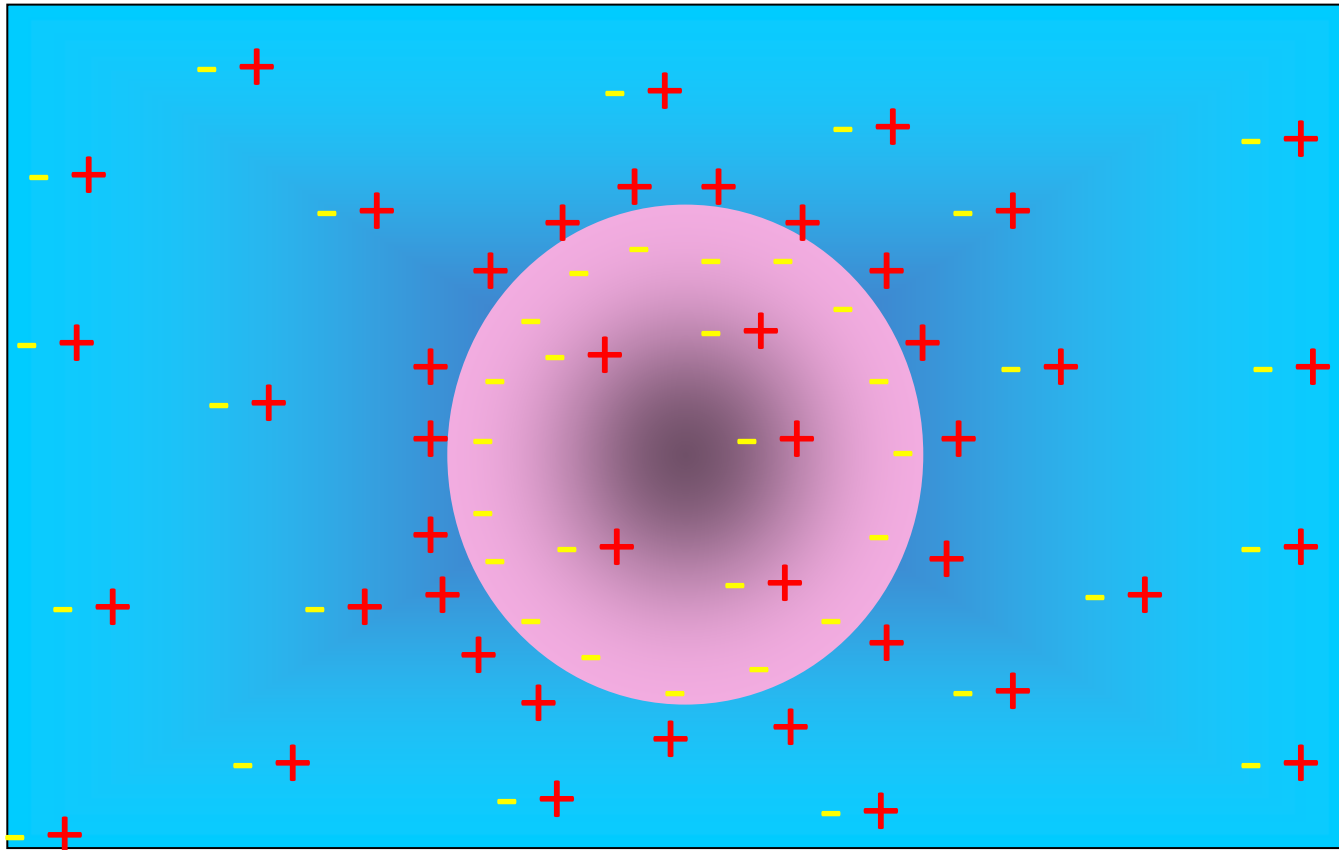
$$\text{current} = \text{will} * \text{way}$$

- Current requires a **pathway** and **motivation** to move
- current = work being done
 - how many charged particles are moving per unit time?
- way = conductance (pathway for ions to move)
 - is there a pathway for the electrons to move?
 - equivalent to open door/ion channels
- will = voltage (driving force)
 - do the ions have a motivation to move?
 - key forces acting on ions: chemical and electrical
 - ions try to move until these forces are balanced

Important Principles

- Ions are either positively or negatively charged
 - Cations are positive
 - Anions are negative
 - Same charges repel each other, opposite charges attract each other
- Charge equivalency: a neutral solution contains equal amounts of positive and negative charges, e.g. for every Na^+ ion there is a partner Cl^- ion.
- When charges move across the membrane they set up a voltage (charge gradient). Only a tiny fraction of all charged ions are needed for this. Hence, although we talk about ion flux, the moving proportion is generally so small that it will not significantly change the concentration of ions inside or outside the cell.
- Charge separation:
 - requires selective movement of specific charge across the membrane, e.g. only Na^+ or only K^+
 - this leads to a charge difference and thus electrical potential (voltage, V) across the membrane
 - requires a pathway for specific charges to move, e.g. selective ion channels

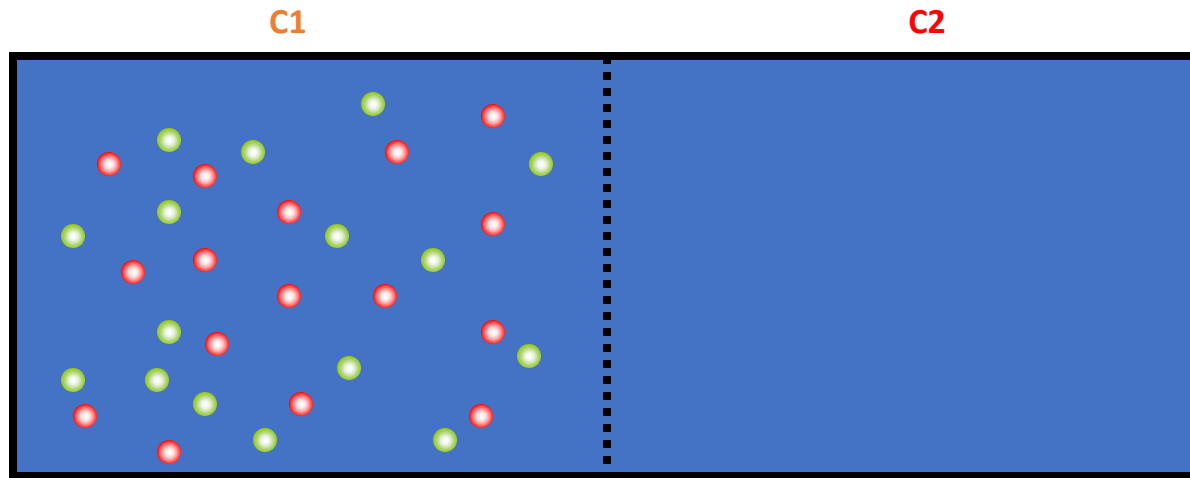
Charge Distribution with Membrane Potential



The resting membrane potential

- What is required?
 - a semi-permeable membrane (blocking water and ion flow)
 - a concentration gradient of ions across the membrane (driven by Na-K ATPase)
 - a pathway for charged ions to flow across the membrane: selective channels
- How is it formed?
 - charge separation across the membrane via selective ion channels
- What is it determined by?
 - which ion conductance is dominant at rest
 - the magnitude of the concentration gradient for that ion
- Measurement and typical values
 - written as inside relative to outside, e.g. -50 mV means interior is 50 mV more negative than exterior
 - usually for an epithelial (non-excitabile) cell -50 mV or more negative, for a neuron (excitable) -80 mV
 - specific value depends largely upon the type of ion channels expressed in a cell

Electrochemical Diffusion

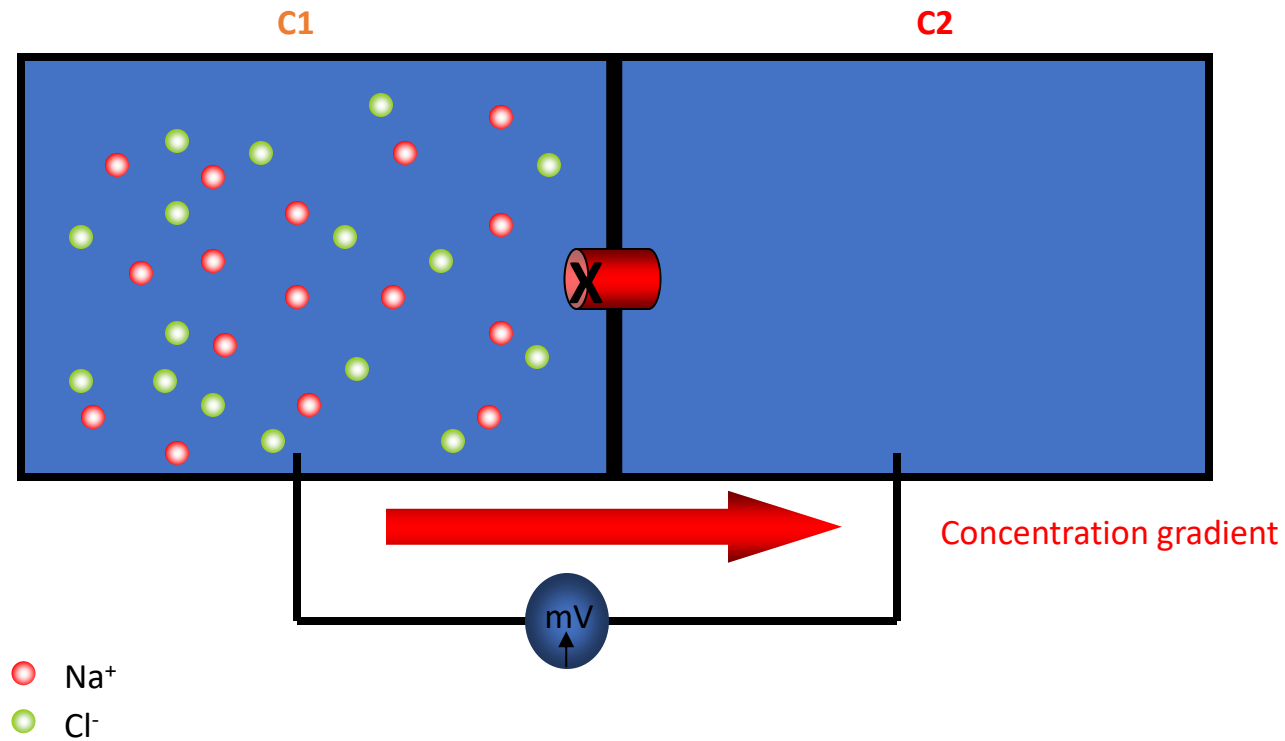


Ionic solutions have electroneutrality (+ = -)

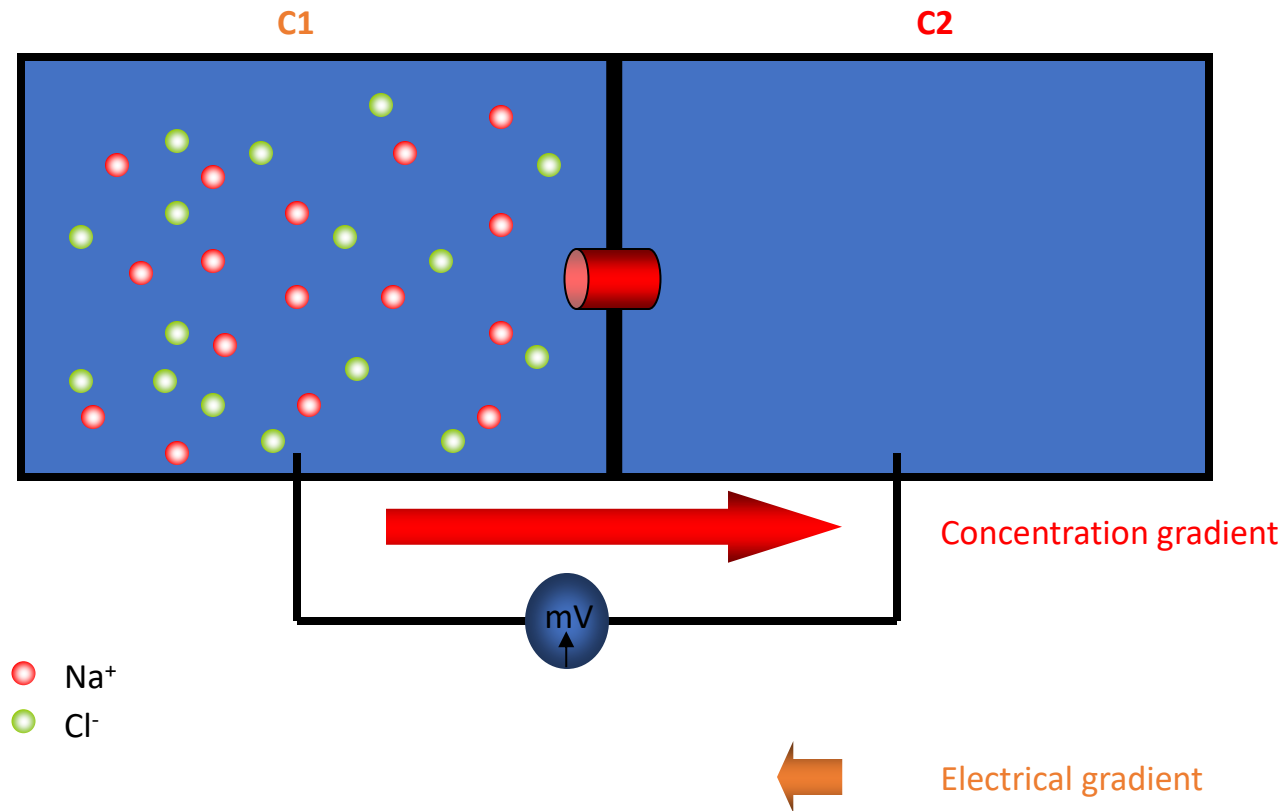


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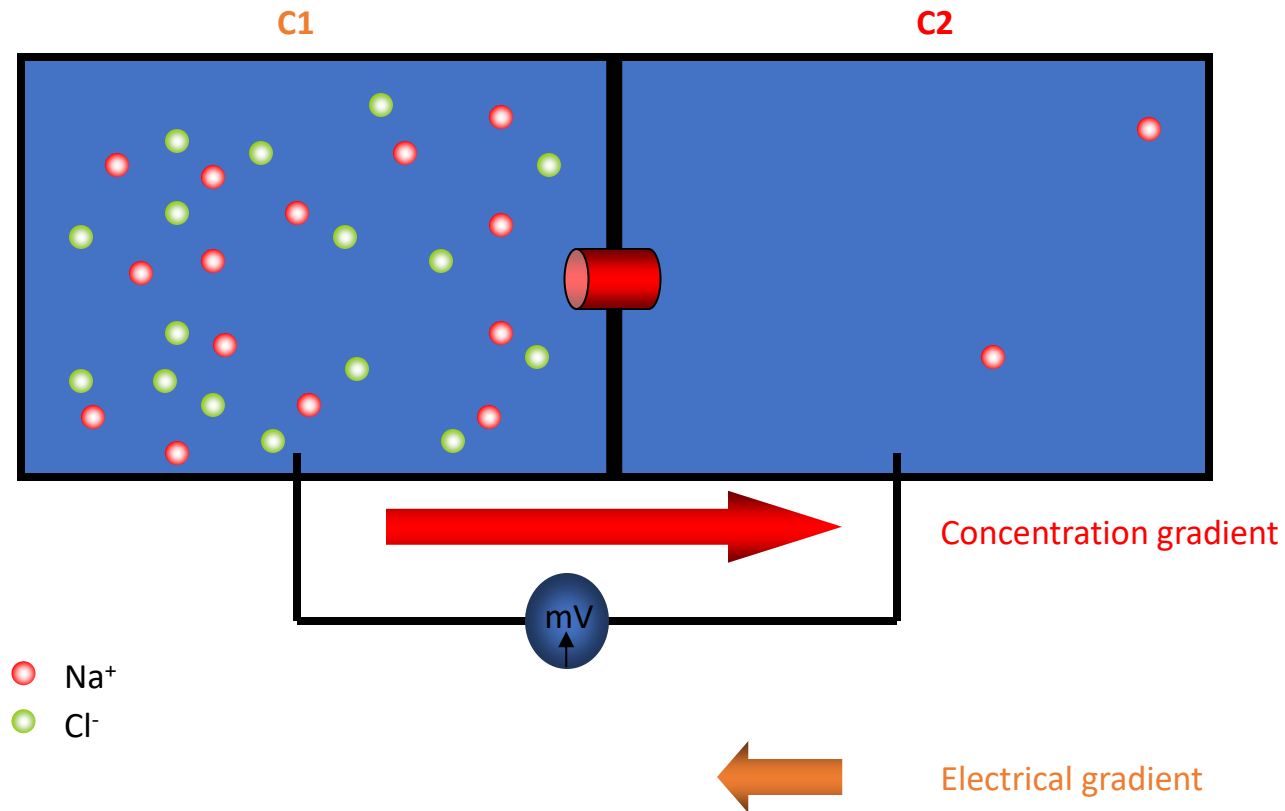
Electrochemical Diffusion



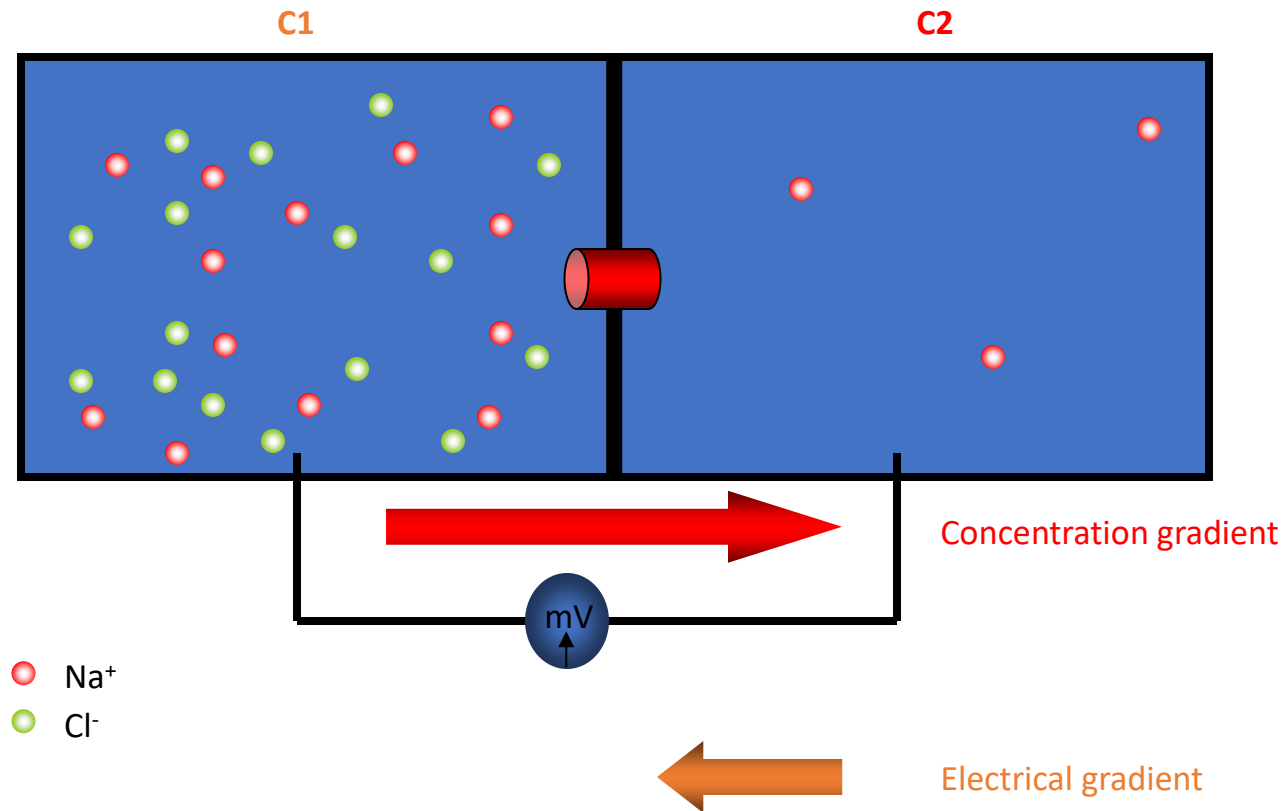
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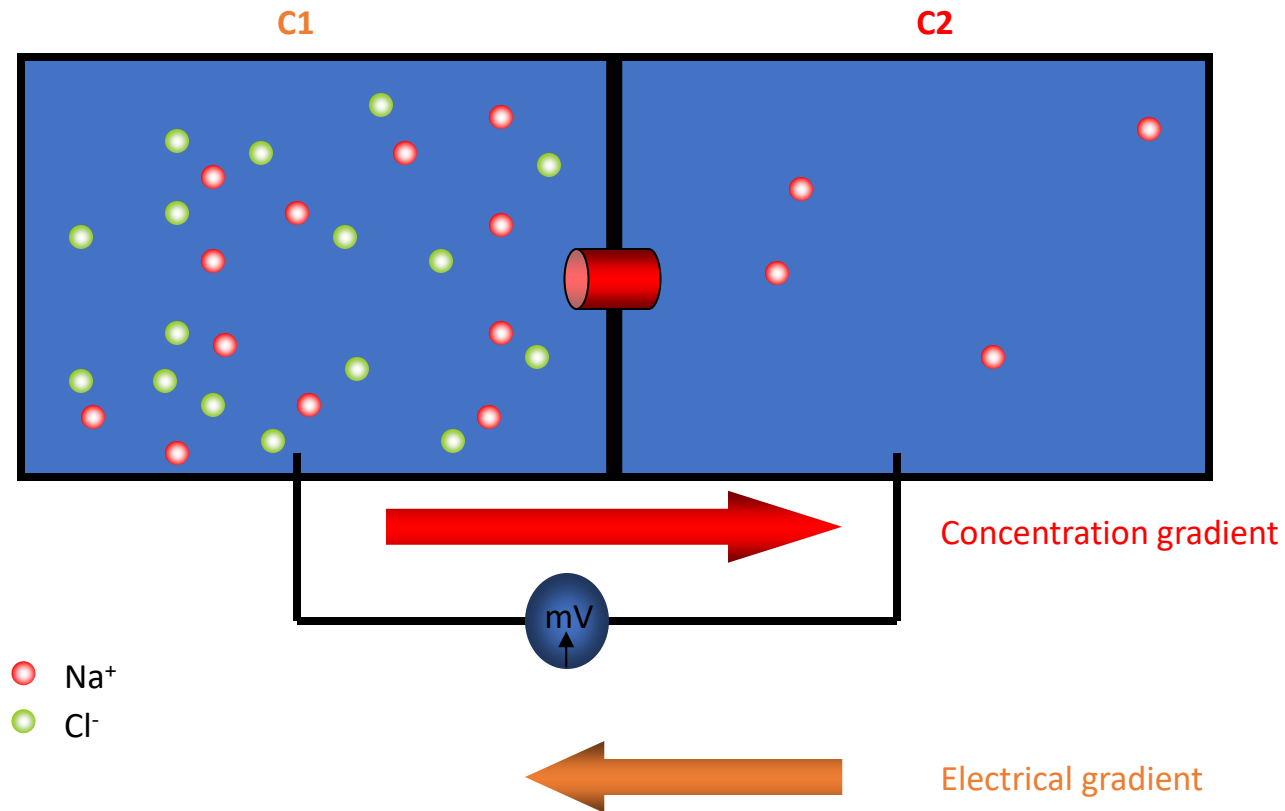
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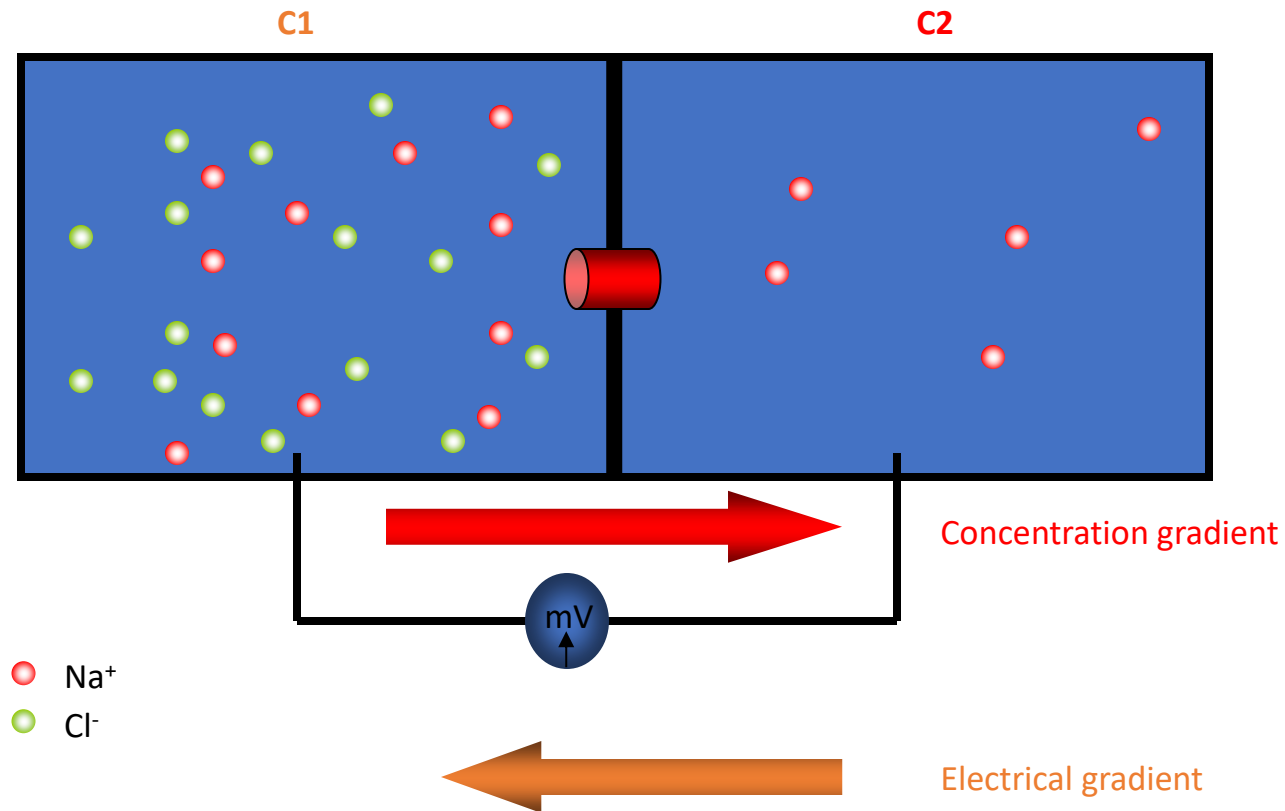
Electrochemical Diffusion



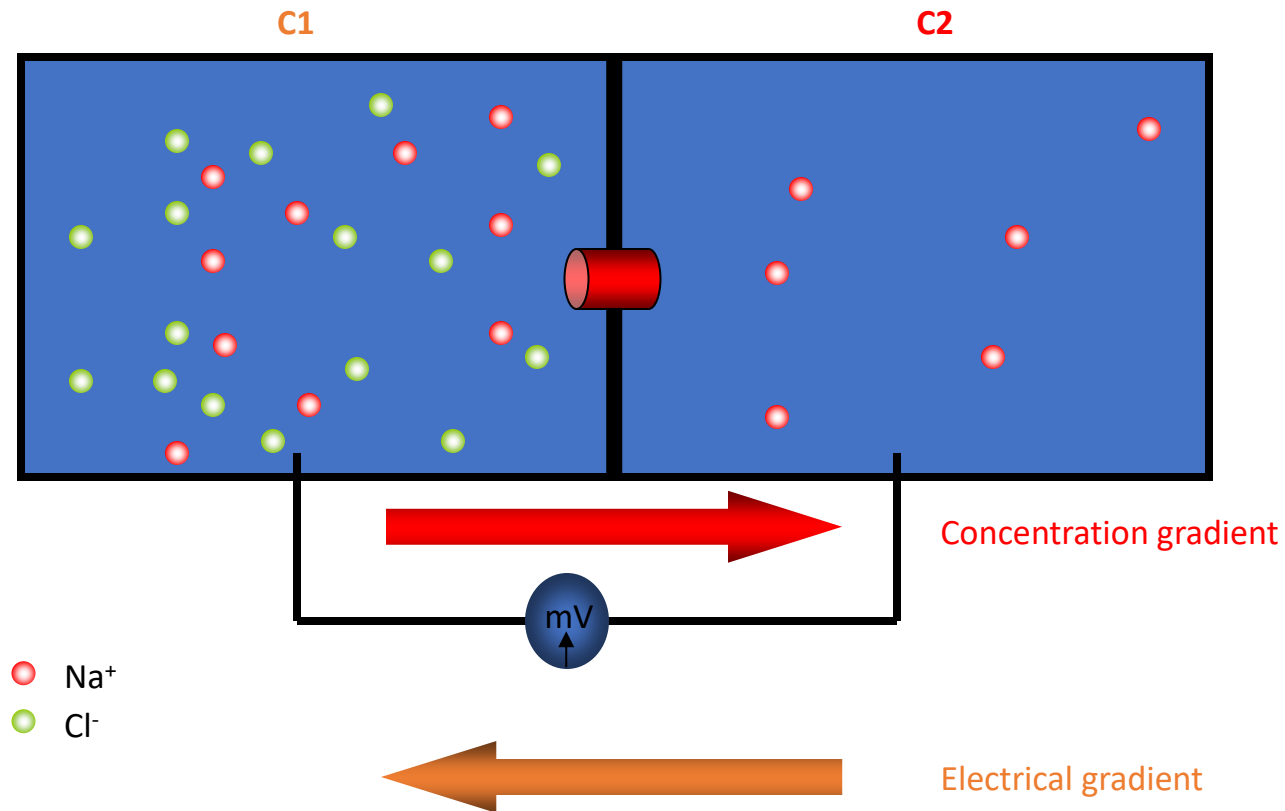
Electrochemical Diffusion



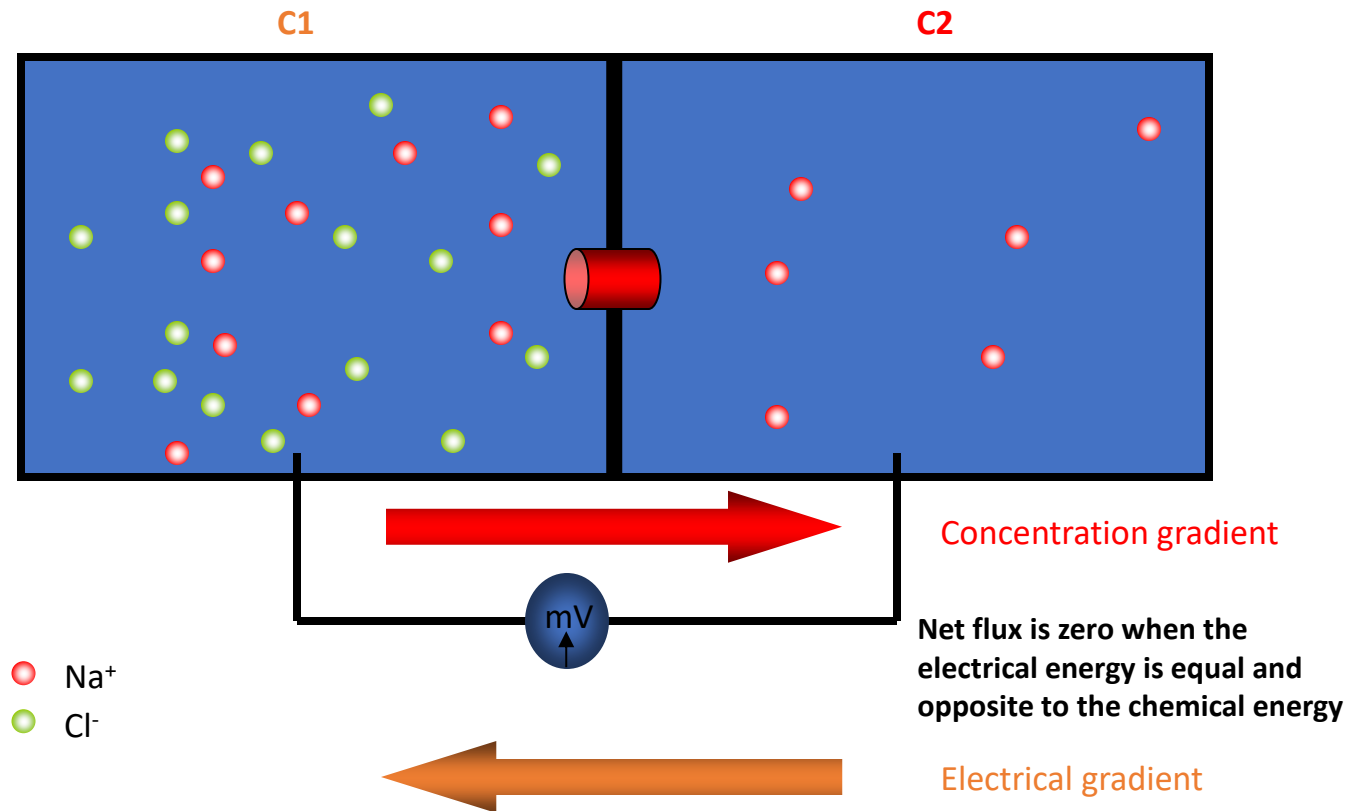
Electrochemical Diffusion



Electrochemical Diffusion



Electrochemical Diffusion

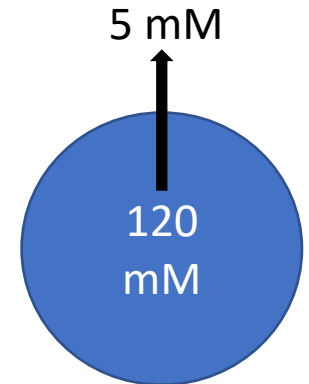


Electrochemical equilibrium potentials

- Steady state model: constant ion pumping (active, e.g. Na-K ATPase) and leakage (passive via channels) results in uneven distribution of key ions across the membrane.
- Net movement of each ion is driven by **chemical gradient** (concentration difference) and **electrical gradient** (charge difference), which we refer to as the combined **electrochemical gradient**.
- Overall net movement of an ion is 0 when electrical and chemical gradient balance each other out. Hence each ion with an uneven chemical gradients across the membrane generates a specific electrical charge gradient.
- This electrical potential is called the **equilibrium potential** for the ion, sometimes also referred to as the Nernst equilibrium potential, since it can be calculated from the chemical gradient via the Nernst equation.

Example for K^+

- An outwardly directed concentration gradient for K^+ results from the ion concentration difference. K^+ ions will flow down that gradient if allowed (Fick's law, diffusion).
- If we open a potassium channel in the membrane, positive K^+ ions leave the cell, resulting in a selective loss of positive charge from the inside (charge separation).
- As a result, the interior of the cells becomes electrically negative.
- That internal negativity is the resting membrane potential!
- The flow of K^+ out of the cell eventually stops because the increasingly negatively charged interior attracts the cationic (positive) K^+ and prevents it leaving the cell.
- A stable equilibrium between chemical and electrical gradient is reached when the electrical gradients and the chemical gradients are equal and opposite.
- Hence the electrical charge gradient is determined by the magnitude of the concentration gradient for the ion. It can be calculated by the Nernst equation.



Key ion species and typical concentrations

Ion type (+/-)	Valency (charge count)	Ion name	Concentration (mM)	
			Intracellular	Extracellular
Cation	Monovalent	Na ⁺	14	140
		K ⁺	120	5
	Bivalent	Ca ²⁺	0.0001	2
Anion	Monovalent	Cl ⁻	8	100

The Nernst Equation

$$E_{ion} = - \frac{RT}{zF} \times \log \left(\frac{\text{inside concentration}}{\text{outside concentration}} \right)$$

- Constants:
 - R = gas constant (8.314 J/(K*mol))
 - T = temperature (in Kelvin, 273 K = 0 °C)
 - z = valency (charge number of the ion in question, no units)
 - F = Faraday constant (96,840 Coulombs/mol)
- At body temperature (37 °C = 310 K): $\frac{RT}{zF} \approx \frac{-60 \text{ mV}}{z}$
- Example for K⁺:
 - $E_K = \frac{-60 \text{ mV}}{+1} \times \log \left(\frac{120}{5} \right) = -60 \text{ mV} \times 1.38 = -82.8 \text{ mV}$
 - i.e. with a K gradient of 120 mM inside to 5 mM outside, and the presence of no other ion conductance, the membrane potential would be -82.8 mV. This membrane potential would lead to a net flux of K⁺ of zero – hence the name equilibrium potential. The inside of the cell has become negative enough to prevent flux of K out of the cell down its chemical gradient.

Ion-specific equilibrium potentials

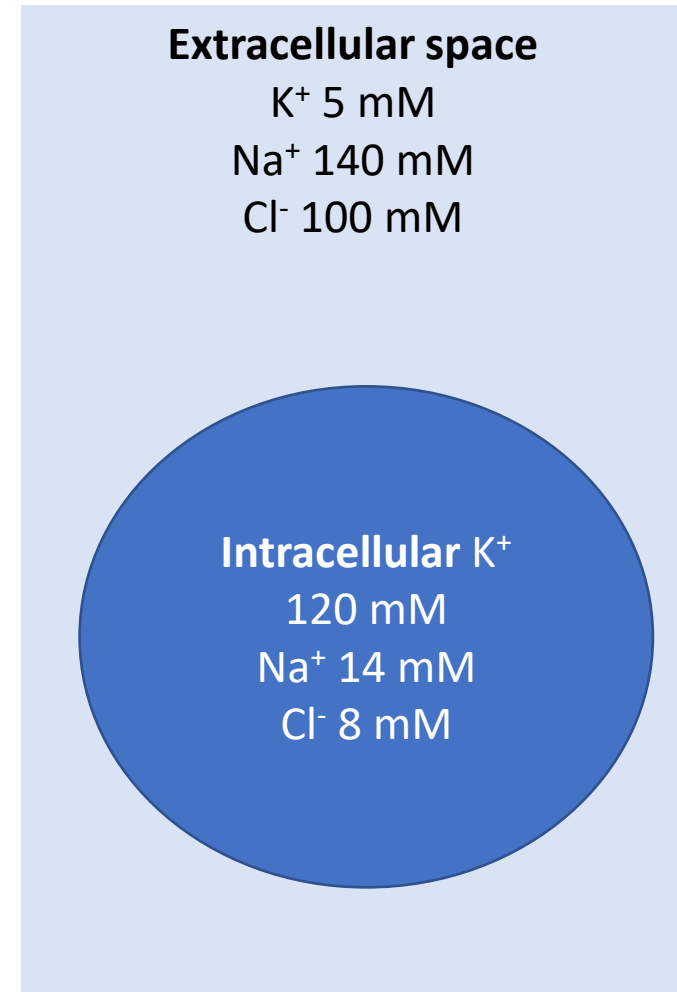
Ion	Concentration (mM)		E_{ion} (mV)
	Intracellular	Extracellular	
Na^+	14	140	$\frac{-60}{+1} \times \log\left(\frac{14}{140}\right) = -60 \times -1 = +60$
K^+	120	5	$\frac{-60}{+1} \times \log\left(\frac{120}{5}\right) = -60 \times 1.38 = -83$
Ca^{2+}	0.0001	2	$\frac{-60}{+2} \times \log\left(\frac{0.0001}{2}\right) = -30 \times -4.3 = +129$
Cl^-	8	100	$\frac{-60}{-1} \times \log\left(\frac{8}{100}\right) = 60 \times -1.1 = -66$

V_m versus E_{ion}

- V_m - membrane potential (measured) is the net effect of all charge movements taking place across the membrane. Most affected by the ion species with the largest conductance across the membrane (and requires a concentration gradient of that ion).
- E_{ion} – the equilibrium potential for a single ion species (e.g. E_{Na} –sodium equilibrium potential). Calculated potential that would be generated across the membrane if this ion species were the only conductance present, determined by the size of the ion gradient.

Mixing equilibrium potentials in a cell

- Implications of Nernst equilibrium potentials:
 - If the cell only had a K^+ conductance:
 $V_m = -82 \text{ mV}$
 - If the cell only had a Na^+ conductance:
 $V_m = +60 \text{ mV}$
 - If the cell only had a Cl^- conductance:
 $V_m = -66 \text{ mV}$
- However, most cells will have all of those conductances
- V_m will be closest to the E_{ion} of the ion with the largest fractional conductance
- All other ion species contribute to V_m according to their **fractional conductance**



Fractional Conductance

- Membrane can dynamically change conductance for specific ions by opening and closing selective channels
- More open channels or more flow per channel →
↑conductance
- The ion with the largest conductance across the membrane has the most influence on the resting membrane potential
 - most cells have a high K permeability at rest
 - hence RMP is close to the K⁺ equilibrium potential in most cells
 - changing extracellular K⁺ concentrations has strongest effect on RMP
- All other ions contribute proportionally to their relative conductance
- Overall V_m for multiple ion species can be calculated via the Goldman-Hodgkin-Katz or chord conductance equation

The Goldman-Hodgkin-Katz equation

$$V_m = -\frac{RT}{F} \ln \left(\frac{P_K[K]_i + P_{Na}[Na]_i + P_{Cl}[Cl]_o}{P_K[K]_o + P_{Na}[Na]_o + P_{Cl}[Cl]_i} \right)$$

P = permeability; g = conductance. This is like a chained Nernst equation for each ion species, weighted by fractional conductance. A simpler way of writing this:

$$V_m = \left(\frac{g_K}{g_{total}} \times E_K \right) + \left(\frac{g_{Na}}{g_{total}} \times E_{Na} \right) + \left(\frac{g_{Cl}}{g_{total}} \times E_{Cl} \right)$$

Can we predict if an ion will flow into or out of a cell?

- Yes – but we need to know a few things.
 1. what is the cells membrane potential (V_m)?
 2. what is the ions calculated equilibrium potential (E_{ion})?
 - If the equilibrium potential is the membrane potential at which net flux is zero – then clearly when V_m and E_{ion} do not match the ion is not in equilibrium and is thus in flux.
 - Therefore, the bigger the difference between V_m and E_{ion} , the bigger the driving force for flux

Driving Force

- If the Nernst potential is the E_{ion} where there is no net flux of the ion – then the further V_m is from the E_{ion} the higher the flux of that ion will be
- Driving Force is the difference between the actual measured V_m and the calculated equilibrium potential for a particular ion

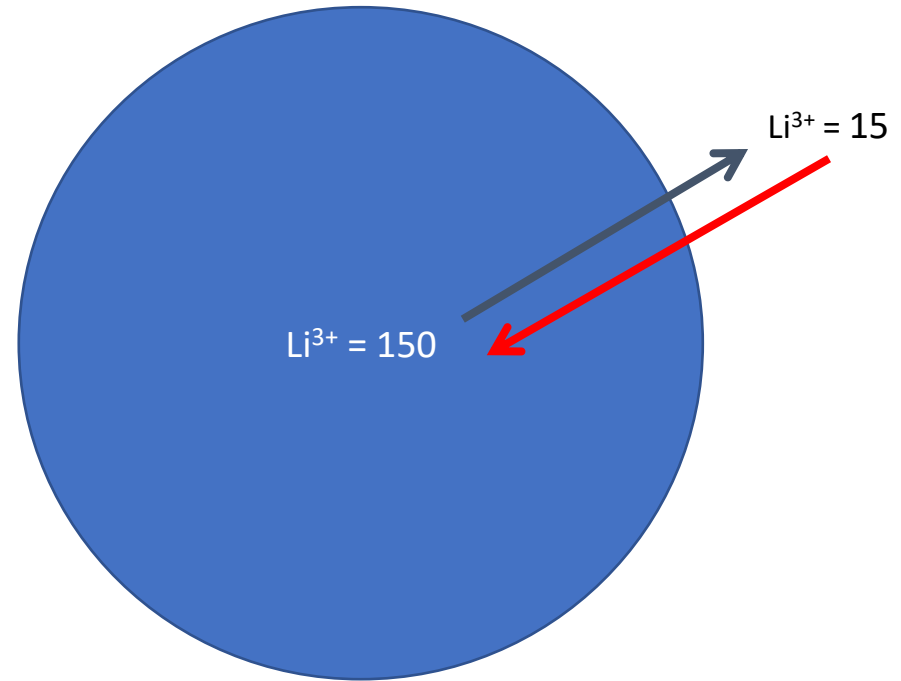
$$DF = V_m - E_{ion}$$

- a positive value means a cationic efflux or an anionic influx, and the opposite for negative DF values:

DF	Cation (+)	Anion (-)
+	efflux	influx
-	influx	efflux

Answer

- $E_{\text{Li}^{3+}} = \frac{-60}{+3} \times \log\left(\frac{150}{15}\right) = -20 \times 1 = -20 \text{ mV}$
- cell interior must be negative to prevent positive charge leak
- At -40 mV V_m is 20 mV more negative than equilibrium
- hence net driving force of 20 mV for Li^{3+} influx



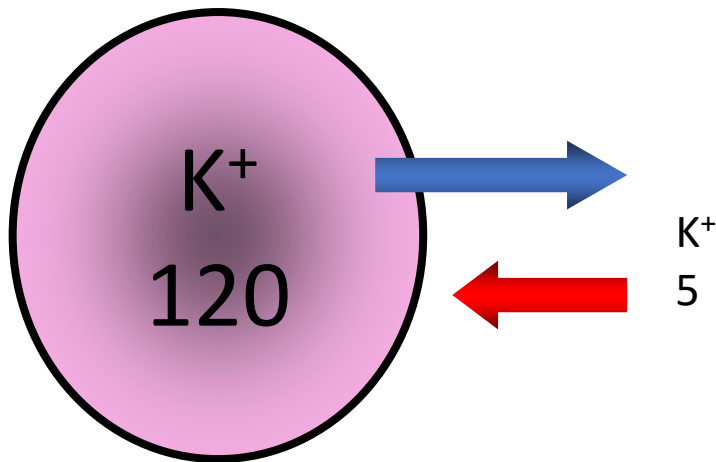
$E_{\text{Li}^{3+}} = -20 \text{ mV}$ (equilibrium)
 $V_m = -40 \text{ mV}$ (more negative)
 $\text{DF} = -20 \text{ mV}$ (net influx)

Summary

- Membrane potential is based on selective ion transport
- The resulting electro-chemical gradient leads to specific **equilibrium potential** for each type of ion
- All ions contribute to V_m **proportionally to their relative conductance**
- **At rest, the dominant K^+ conductance** leads to a negative V_m of -50 to -80 mV for most cells
- Outlook for next lecture: An action potential is simply a transient, dynamic change in conductance from the dominant E_K (-80 mV) at rest to E_{Na} (+60 mV)
- Please go over DLA 4 'Epithelial transport' before SG3

Self study: Driving force & flux examples

E_K and V_m



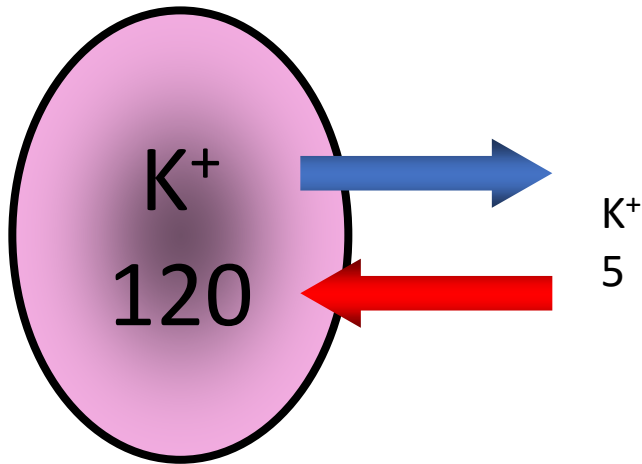
$$DF = V_m - E_{ion}$$

$$V_m = -50 \text{ mV}$$

$$DF = (-50) - (-83) = +33$$

Outward net flux

E_K and V_m



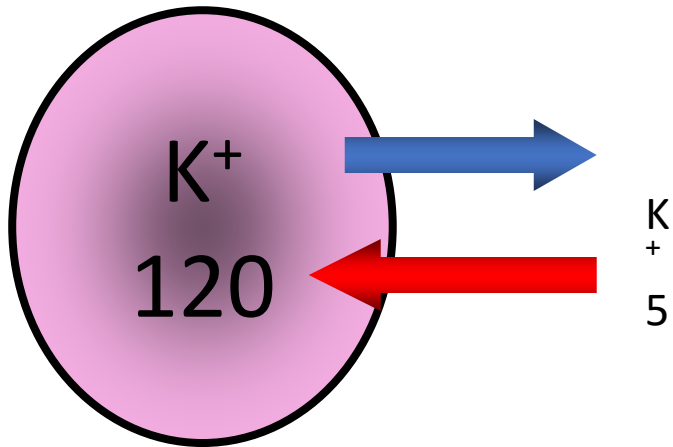
$$DF = V_m - E_{ion}$$

V_m of -83 mV
Means no net flux

$$DF = (-83) - (-83) = 0$$

No driving force

E_K and V_m



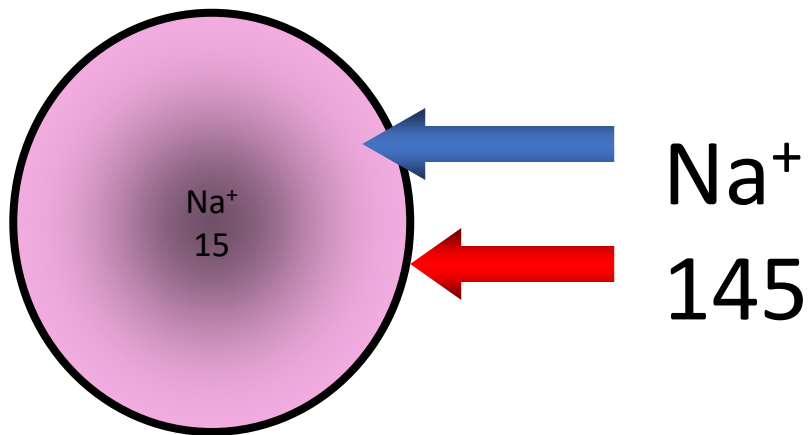
$$DF = V_m - E_{ion}$$

$$V_m = -95 \text{ mV}$$

$$DF = (-95) - (-83) = -12$$

Inward net flux

E_{Na} and V_m



$$DF = V_m - E_{ion}$$

V_m of -50 mV

$$DF = (-50) - (+59) = -109$$

Inward net flux

This is the reason
that many transport systems
use Na as a co- or anti-
port partner – there is a lot
of energy for Na influx!